Guillain-Barre Syndrome

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Guillain-Barré Syndrome (GBS)

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Introduction

Guillain-Barré syndrome is a collection of clinical syndromes manifesting as an **acute inflammatory polyradiculoneuropathy** with resultant **weakness and diminished reflexes**.

Other names: Landry's ascending paralysis (weakness starts in the legs and moves up toward trunk and face); acute idiopathic polyneuritis; acute polyradiculoneuritis; Guillain-Barré-Strohl syndrome.

> **Classic description:** a **demyelinating neuropathy with ascending weakness**, characterised by **weakness, paraesthesias and hyporeflexia.** It is a **post-infectious** polyneuropathy causing demyelination of **mainly motor** (sometimes also sensory) nerves. In severe cases, muscle weakness leads to **respiratory failure**, and **labile autonomic dysfunction** may occur.

History: In 1859 **Landry** reported 10 patients with ascending paralysis. In 1916, **Guillain, Barré and Strohl** described 2 French soldiers with motor weakness, areflexia, **CSF albuminocytologic dissociation**, and diminished deep tendon reflexes.

AIDP (acute inflammatory demyelinating polyradiculoneuropathy) is the most widely recognised form. *Poly* = many nerves; *radiculo* = the nerve roots (where nerves exit the spinal cord); *neuropathy* = disease/damage of nerve tissue.

Variants: Acute motor axonal neuropathy (**AMAN**); acute motor-sensory axonal neuropathy (**AMSAN**); **Miller-Fisher syndrome**.

Epidemiology

- Affects **all ages but uncommon in children**.
- **Male:female 1.5:1**.
- The **most common neuromuscular disorder in the developed world** (unlike polio in the developing world).

Pathophysiology

Characterised by **absent or profoundly delayed conduction** in nerve fibres, from **demyelination of peripheral nerves and spinal roots**. **Cranial nerves are rarely involved.**

> GBS is an **autoimmune response triggered by an antecedent viral infection**, with both **humoral and cell-mediated** components. Symptoms usually result from **immune-mediated injury to the myelin sheath**; in a few cases **axonal damage** results from a direct cellular immune attack on the axon itself.

Aetiology

- **Two-thirds** have a preceding **GI or respiratory infection 1-3 weeks** before the weakness.
- **Infections:** EBV, CMV, Chlamydia, HBV, **Campylobacter**, Mycoplasma, HIV.
- **Vaccinations:** rabies, streptococcus, influenza.
- **Malignancies:** lymphomas, particularly **Hodgkin disease**.
- **Drugs:** captopril, gold, penicillamine.

Clinical features

Typically presents **2-4 weeks after a benign respiratory or GI illness**. Weakness may progress over **hours to days** to involve arms, trunk, cranial nerves and muscles of respiration.

- **Sensory:** most complain of **paraesthesias/numbness**, beginning in the **toes and fingertips** and progressing up, generally **not beyond the wrists or ankles**. **Pain** is aching/throbbing, worst in the **shoulder girdle, back, buttocks and thighs**, provoked by the slightest movement.
- **Motor:** rapidly progressive weakness usually **starts in the lower limbs and ascends** to the upper limbs, respiratory and facial muscles. **Bilateral and symmetrical.** Cranial nerve involvement -> facial weakness, dysphagia, drooling, dysarthria. In the **Miller-Fisher variant**, neuropathy **begins with cranial nerve deficits**. Usually **maximal in severity ~2 weeks** after onset.
- **Autonomic:** tachycardia, bradycardia, facial flushing, paroxysmal hypertension, orthostatic hypotension, anhidrosis/diaphoresis, urinary retention.
- **Respiratory:** dyspnoea on exertion, shortness of breath, difficulty swallowing, slurred speech. **Ventilatory failure requiring support occurs in up to one-third** of patients at some point.
- **Signs:** flaccid paralysis with **lost deep tendon reflexes**; **hypotonia, hyporeflexia/areflexia and sensory loss.**

Clinical course

Usually **benign**. Spontaneous recovery begins within **2-3 weeks**. Most regain full strength, though some have residual weakness.

> Recovery follows a gradient **inverse to the direction of involvement** – **bulbar function recovers first, lower-limb weakness resolves last**, and **tendon reflexes are usually the last to recover.** Bulbar and respiratory involvement may lead to **death if treatment is delayed.**

Diagnosis

- **CSF analysis:** elevated CSF protein ($>$ **400 mg/L**) with **no rise in cell count** – **cyto-albuminologic (albuminocytologic) dissociation**. Protein may be **normal very early**.
- **Serology:** four-fold rise in serum antibodies against microbes.
- **EMG:** acute denervation of muscle; **weak muscles show reduced recruitment**.
- **Serum CK:** mildly elevated or normal. **Muscle biopsy** not usually required.
- **Nerve conduction studies:** slowing **2-3 weeks** after onset – signs of demyelination: **conduction slowing, prolonged distal latencies, prolonged F-waves, conduction block/dispersion**.
- **ECG:** may show conduction/rhythm disorders.
- **Lung function:** forced vital capacity in respiratory involvement.

> **Negative inspiratory force (NIF)** is an easy bedside test of respiratory muscle function; **normal is $>$ 60 cm H O.** If NIF is **dropping or nears 20 cm H O**, respiratory support must be available.

Differentials & complications

Differentials: poliomyelitis, hypokalaemia, botulism, cauda equina syndrome, diphtheria, myasthenia gravis, spinal cord injury, transverse myelitis, organophosphate poisoning, hereditary neuropathies, porphyria.

Complications: respiratory failure, hypotension/hypertension, thromboembolism (DVT), hypostatic pneumonia, decubitus ulcers, cardiac arrhythmia, ileus, aspiration, urinary retention.

Treatment

- About **one-third require ICU admission**, mainly for **respiratory failure**. Consider ICU for **labile dysautonomia, FVC $<$ 20 mL/kg, or severe bulbar palsy**, or any sign of respiratory compromise.
- **Specific therapy: plasma exchange** or **IV immunoglobulin (IVIG)** – shown to **shorten recovery time by as much as 50%.**
- **Prednisolone 2 mg/kg/day is ineffective by itself** (may be given alongside plasmapheresis).
- **DVT prophylaxis** for the paralysed extremities.
- **Autonomic management:** atropine for symptomatic bradycardia; short-acting agents (beta-blocker or nitroprusside) for hypertension; IV fluids and supine positioning for hypotension. Tachycardia rarely needs treatment.
- **Supportive/rehabilitation:** respiratory support, cardiac monitoring, safe nutrition, monitoring for infection; **physical and occupational therapy** to regain strength, mobility, coordination and independence; pain management; regular monitoring (some relapse).

References

Nelson Textbook of Paediatrics; Azubuike & Nkanginieme, Paediatrics and Child Health in a Tropical Region; lecture material of Dr Chidomere R.I.